

Lie Groups, Lie Algebras, and Representations

Reading Notes and Exercises

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1 Matrix Lie Groups

1.1 Definitions

Definition 1.1. The **general linear group** over the real numbers, denoted $GL(n, \mathbb{R})$, is the group of all $n \times n$ invertible matrices with real entries. The general linear group over the complex numbers, denoted $GL(n, \mathbb{C})$, is the group of all $n \times n$ invertible matrices with complex entries.

Definition 1.2. Let $M_n(\mathbb{C})$ denote the space of all $n \times n$ matrices with complex entries.

Definition 1.3. Let A_m be a sequence of complex matrices in $M_n(\mathbb{C})$. We say A_m **converges** to a matrix A if each entry of A_m converges to the corresponding entry of A as $m \rightarrow \infty$, i.e. if $(A_m)_{jk}$ converges to A_{jk} for all $1 \leq j, k \leq n$.

Note that Definition 1.3 means we can identify $M_n(\mathbb{C})$ with $\mathbb{C}^{n^2}$, and use the standard notion of convergence in $\mathbb{C}^{n^2}$. All topological properties of a matrix Lie group G will be considered with respect to the topology G inherits as a subset of $M_n(\mathbb{C}) \cong \mathbb{C}^{n^2}$.

We now consider subgroups of $GL(n, \mathbb{C})$, which are subsets $G \subset GL(n, \mathbb{C})$ such that $Id \in G$, and for all $A, B \in G$ we have $AB \in G$ and $A^{-1} \in G$.

Definition 1.4. A **matrix Lie group** is a subgroup G of $GL(n, \mathbb{C})$ with the property: if A_m is any sequence of matrices in G , and A_m converges to some matrix A , then either A is in G or A is not invertible.

This property means that G is a closed subset of $GL(n, \mathbb{C})$. So this definition is equivalent to saying a matrix Lie group is a closed subgroup of $GL(n, \mathbb{C})$. However, G is not necessarily closed in $M_n(\mathbb{C})$. For this to be true, we require a stronger property: if A_m is any sequence of matrices in G and A_m converges to some matrix A , then $A \in G$.

1.2 Examples

We introduce some important examples of (matrix) Lie groups, including the classical groups.

1.2.1 General and Special Linear Groups

The general linear groups $GL(n, \mathbb{R})$ and $GL(n, \mathbb{C})$, defined in 1.1, are matrix Lie groups.

The special linear groups $SL(n, \mathbb{R})$ and $SL(n, \mathbb{C})$ is the group of $n \times n$ invertible matrices (with real or complex entries) having determinant 1. They are matrix Lie groups.

1.2.2 Unitary and Orthogonal Groups

A $n \times n$ complex matrix A is **unitary** if its column vectors are orthonormal, that is,

$$\sum_{l=1}^n \bar{A}_{lj} A_{lk} = \delta_{jk}$$

where δ_{jk} is the Kronecker delta. This can be rewritten as

$$\sum_{l=1}^n (A^*)_{jl} A_{lk} = \delta_{jk}$$

where A^* is the adjoint of A , defined by $A^* = \bar{A}^t$. A is unitary if and only if $A^* = A^{-1}$.

The **unitary group** $U(n)$ is the collection of all unitary matrices. The **special unitary group** $SU(n)$ is the subgroup of $U(n)$ consisting of unitary matrices with determinant 1. $U(n)$ and $SU(n)$ are matrix Lie groups.

A $n \times n$ real matrix A is **orthogonal** if its column vectors are orthonormal. The equivalent conditions are the same as before, except that when A is real, $A^* = A^t$. A is orthogonal if and only if $A^t = A^{-1}$, if and only if A preserves the inner product on $\mathbb{R}^n$.

The **orthogonal group** $O(n)$ is the collection of all orthogonal matrices. The **special orthogonal group** $SO(n)$ is the set of $n \times n$ orthogonal matrices with determinant 1. Elements of $SO(n)$ are rotations, while the elements of $O(n)$ are either rotations or combinations of rotations and reflections.

Consider the bilinear form $(\cdot, \cdot)$ on $\mathbb{C}^n$ defined by

$$(x, y) = \sum_j x_j y_j$$

Note that this form is not an inner product, since it is symmetric, not conjugate-symmetric. The **complex orthogonal group** $O(n, \mathbb{C})$ is the set of all $n \times n$ complex matrices A that preserve this form. The group $SO(n, \mathbb{C})$ is the set of all matrices in $O(n, \mathbb{C})$ with determinant 1. $O(n, \mathbb{C})$ and $SO(n, \mathbb{C})$ are matrix Lie groups.

1.2.3 Generalized Orthogonal and Lorentz Groups

Consider $\mathbb{R}^{n+k}$, with n and k positive integers. Define a symmetric bilinear form $[\cdot, \cdot]_{n,k}$ on $\mathbb{R}^{n+k}$ by

$$[x, y]_{n,k} = x_1 y_1 + \cdots + x_n y_n - x_{n+1} y_{n+1} - \cdots - x_{n+k} y_{n+k}$$

The **generalized orthogonal group** $O(n, k)$ is the set of $(n+k) \times (n+k)$ real matrices A that preserve this form, i.e. $[Ax, Ay]_{n,k} = [x, y]_{n,k}$. It is a matrix Lie group. In particular, the **Lorentz group** is $O(3, 1)$. $SO(n, k)$ is the subgroup of $O(n, k)$ consisting of elements with determinant 1.

For any $A \in O(n, k)$, we have $\det A = \pm 1$.

1.2.4 Symplectic Groups

Define the skew-symmetric bilinear form ω on $\mathbb{R}^{2n}$ by

$$\omega(x, y) = \sum_{j=1}^n (x_j y_{n+j} - x_{n+j} y_j)$$

The **real symplectic group** $Sp(n, \mathbb{R})$ is the set of all $2n \times 2n$ real matrices A which preserve ω . It is a closed subgroup of $GL(2n, \mathbb{R})$, hence a matrix Lie group. $\det A = \pm 1$ for all $A \in Sp(n, \mathbb{R})$; in fact, $\det A = 1$.

Using the same formula, we can define a bilinear form ω on $\mathbb{C}^{2n}$. Over $\mathbb{C}$, we have the relation $\omega(z, w) = (z, \Omega w)$, where $(\cdot, \cdot)$ is the complex bilinear form from 1.2.2. The **complex symplectic group** $Sp(n, \mathbb{C})$ is the set of all $2n \times 2n$ complex matrices which preserve this form. We can show that for all $A \in Sp(n, \mathbb{C})$, $\det A = \pm 1$, and in fact, $\det A = 1$.

The **compact symplectic group** $Sp(n)$ is defined as

$$Sp(n) = Sp(n, \mathbb{C}) \cap U(2n)$$

That is to say, $Sp(n)$ is the group of $2n \times 2n$ matrices that preserve both the inner product and the bilinear form ω .

1.2.5 The Euclidean and Poincaré Groups

The **Euclidean group** $E(n)$ is the group of all transformations of $\mathbb{R}^n$ that can be expressed as a composition of a translation and an orthogonal linear transformation. Elements of $E(n)$ are written as pairs $\{x, R\}$ with $x \in \mathbb{R}^n$ and $R \in O(n)$, and $\{x, R\}$ acts on $\mathbb{R}^n$ by

$$\{x, R\}y = Ry + x$$

The product operation for $E(n)$ is

$$\{x_1, R_1\}\{x_2, R_2\} = \{x_1 + R_1x_2, R_1R_2\}$$

The inverse of an element of $E(n)$ is

$$\{x, R\}^{-1} = \{-R^{-1}x, R^{-1}\}$$

$E(n)$ is not a subgroup of $GL(n, \mathbb{R})$, but it is isomorphic to a closed subgroup of $GL(n+1, \mathbb{R})$.

The **Poincaré group** $P(n, 1)$ is the group of all transformations of $\mathbb{R}^{n+1}$ of the form

$$T = T_x A$$

with $x \in \mathbb{R}^{n+1}$ and $A \in O(n, 1)$.

1.2.6 The Heisenberg Group

The **Heisenberg group** H is the set of all 3×3 real matrices

$$A = \begin{pmatrix} 1 & a & b \\ 0 & 1 & c \\ 0 & 0 & 1 \end{pmatrix}$$

where $a, b, c \in \mathbb{R}$. H is a subgroup of $GL(3, \mathbb{R})$.

1.2.7 The Groups $\mathbb{R}^*$, $\mathbb{C}^*$, S^1 , $\mathbb{R}$, and $\mathbb{R}^n$

The group $\mathbb{R}^*$ of non-zero real numbers under multiplication is isomorphic to $GL(1, \mathbb{R})$. The group $\mathbb{C}^*$ of non-zero complex numbers under multiplication is isomorphic to $GL(1, \mathbb{C})$. The group S^1 of complex numbers with absolute value 1 is isomorphic to $U(1)$. The group $\mathbb{R}$ under addition is isomorphic to $GL(1, \mathbb{R})^+$, the 1×1 matrices with positive determinant. The group $\mathbb{R}^n$ under vector addition is isomorphic to the group of diagonal real matrices with positive diagonal entries.

These groups are not defined as groups of matrices, but can be thought of as such.

1.2.8 The Compact Symplectic Group

We try to understand the structure of $Sp(n)$, and show that it can be understood as the "unitary group over the quaternions".

Define a conjugate-linear map $J : \mathbb{C}^{2n} \rightarrow \mathbb{C}^{2n}$ by

$$J(\alpha, \beta) = (-\bar{\beta}, \bar{\alpha})$$

where $\alpha, \beta \in \mathbb{C}^n$ and $(\alpha, \beta) \in \mathbb{C}^{2n}$. For all $z, w \in \mathbb{C}^{2n}$, we have

$$\omega(z, w) = \langle Jz, w \rangle$$

Furthermore, for all $z, w \in \mathbb{C}^{2n}$, we have

$$\langle Jz, w \rangle = -\langle Jw, z \rangle$$

and

$$J^2 = -I$$

Proposition 1.1. *If $U \in U(2n)$, then $U \in Sp(n)$ if and only if U commutes with J .*

Theorem 1.1. *$U \in Sp(n)$ if and only if there exists an orthonormal basis $u_1, \dots, u_n, v_1, \dots, v_n$ for $\mathbb{C}^{2n}$ such that the following properties hold:*

1. $Ju_j = v_j$

2. *For some real numbers $\theta_1, \dots, \theta_n$, we have*

$$\begin{aligned} Uu_j &= e^{i\theta_j}u_j \\ Uv_j &= e^{-i\theta_j}v_j \end{aligned}$$

3. *We have*

$$\begin{aligned} \omega(u_j, u_k) &= \omega(v_j, v_k) = 0 \\ \omega(u_j, v_k) &= \delta_{jk} \end{aligned}$$

Lemma 1.1. *Suppose V is a complex subspace of $\mathbb{C}^{2n}$ that is invariant under the conjugate-linear map J . Then the orthogonal complement $V^\perp$ of V (with respect to the inner product $\langle \cdot, \cdot \rangle$) is also invariant under J . Furthermore, V and $V^\perp$ are orthogonal with respect to ω , i.e.*

$$\omega(z, w) = 0$$

for all $z \in V$ and $w \in V^\perp$.

1.3 Topological Properties

1.3.1 Compactness

Definition 1.5. *A matrix Lie group $G \subset GL(n, \mathbb{C})$ is **compact** if it is compact in the usual topological sense as a subset of $M_n(\mathbb{C}) \cong \mathbb{R}^{2n^2}$.*

By the Heine-Borel theorem, a matrix Lie group G is compact if and only if it is closed and bounded.

$O(n)$, $SO(n)$, $U(n)$, $SU(n)$, and $Sp(n)$ are compact.

1.3.2 Connectedness

Definition 1.6. *A matrix Lie group G is **connected** if for all $A, B \in G$, there exists a continuous path $A(t)$, $a \leq t \leq b$, lying in G with $A(a) = A$ and $A(b) = B$. For any matrix Lie group G , the **identity component** of G , denoted G_0 , is the set of $A \in G$ for which there exists a continuous path $A(t)$, $a \leq t \leq b$, lying in G with $A(a) = I$ and $A(b) = A$.*

This definition is actually called **path connected** in topology. A matrix Lie group is connected if and only if it is path-connected.

To show that a matrix Lie group G is connected, it suffices to show that each $A \in G$ can be connected to the identity by a continuous path lying in G .

Proposition 1.2. *If G is a matrix Lie group, the identity component G_0 of G is a normal subgroup of G .*

G_0 is closed, and hence a matrix Lie group.

Proposition 1.3. *$GL(n, \mathbb{C})$ is connected for all $n \geq 1$.*

Proposition 1.4. *$SL(n, \mathbb{C})$ is connected for all $n \geq 1$.*

Proposition 1.5. *$U(n)$, $SU(n)$, and $SO(n)$ are connected for all $n \geq 1$.*

1.3.3 The Topology of $SO(3)$

Definition 1.7. The **real projective space** of dimension n , denoted $\mathbb{R}P^n$, is the set of lines through the origin in $\mathbb{R}^{n+1}$. Since each line through the origin intersects the unit sphere exactly twice, we may think of $\mathbb{R}P^n$ as the unit sphere S^n with "antipodal" points u and $-u$ identified.

Points in $\mathbb{R}P^n$ can be thought of as pairs $\{u, -u\}$ with $u \in S^n$. There is a natural map $\pi : S^n \rightarrow \mathbb{R}P^n$

$$\pi(u) = \{u, -u\}$$

A distance function can be defined on $\mathbb{R}P^n$ by

$$d(\{u, -u\}, \{v, -v\}) = \min(d(u, v), d(u, -v))$$

The space $\mathbb{R}P^n$ is the upper hemisphere in S^n with antipodal points on the equator identified. $\mathbb{R}P^n$ is the closed unit ball $B^n \subset \mathbb{R}^n$, with antipodal points on the boundary of B^n identified.

Proposition 1.6. There is a continuous bijection between $SO(3)$ and $\mathbb{R}P^3$.

$\mathbb{R}P^n$ is not simply connected, and hence $SO(3)$ is not simply connected.

1.4 Homomorphisms

Definition 1.8. Let G and H be matrix Lie groups. A map Φ from G to H is called a **Lie group homomorphism** if

1. Φ is a group homomorphism, and
2. Φ is continuous.

If, in addition, Φ is one-to-one and onto and the inverse map Φ^{-1} is continuous, then Φ is called a **Lie group isomorphism**.

If G and H are matrix Lie groups, and there exists a Lie group isomorphism from G to H , then G and H are isomorphic, $G \cong H$.

The relationship between $SU(2)$ and $SO(3)$: they are not isomorphic, but there exists a Lie group homomorphism $\Phi : SU(2) \rightarrow SO(3)$ that is two-to-one and onto.

Proposition 1.7. The map $U \mapsto \Phi_U$ is a 2-1 and onto map of $SU(2)$ to $SO(3)$, with kernel equal to $\{I, -I\}$.

$SU(2)$ is homeomorphic to S^3 . The proposition shows that $SO(3)$ is homeomorphic to $\mathbb{R}P^3$, that is, S^3 with antipodal points identified.

1.5 Lie Groups

The text first gives a very brief discussion of manifolds and smooth manifolds.

Definition 1.9. A **Lie group** is a smooth manifold G which is also a group, and such that the group product $G \times G \rightarrow G$ and the inverse map $G \rightarrow G$ are smooth.

Every matrix Lie group is a Lie group, but the reverse is not true. However, we focus on matrix Lie groups.

Usually, we call a map Φ between two Lie groups a Lie group homomorphism if Φ is a group homomorphism and Φ is smooth.

Every matrix Lie group G is a manifold, so G must be locally path connected, so G is connected if and only if it is path connected.

1.6 Exercises